Overview

- Composting: is an alternative to deep burial as a biosecure and environmentally sound method of disposal. Further evaluation of this method is required to ensure it is practical and operational in a large animal disease response.
- Pyrolysis: Pyrolysis has the potential to be used as another alternative method of carcass disposal and safely inactivates pathogens during large animal disease responses.

Project Lead: New South Wales Department of Primary Industries and Regional Development (NSW DPIRD)

Project Contact: Animal.biosecurity@dpi.nsw.gov.au

Project Status: Complete

Objective alignment:

- 1 – Improve Australia's preparedness and ability to respond to emergency animal diseases.

Activity alignment:

- 1.3 – Operationalise AUSVETPLAN manuals and documents across industry supply chains and structures.
- 1.4 – Undertake projects, including commissioning and undertaking research, to further develop economic analyses and epidemiological modelling tools supporting rapid decision-making in emergency animal disease responses for priority diseases.

Project updates

Future Actions

Composting trials have been completed, with the final report in progress.

Development of an operational manual is planned, based on the findings from the trials.

November 2025

Composting: This project has successfully demonstrated that composting cattle and pig carcasses following mass mortality events is both feasible and effective under Australian conditions. The trials provided strong evidence that well-designed and managed windrows can achieve the required pathogen reduction standards and produce material suitable for beneficial land application. The project will build confidence among regulators, industry stakeholders and operational staff that composting can serve as a practical, biosecure and environmentally sound option for carcass disposal in emergency situations.

November 2024

Pyrolysis: The pyrolysis implementation project is currently still in the build phase of the pyrolysis unit. There have been several delays in sourcing materials, some labour issues and the need for design modifications that have led to this delay. Once a new handover date for the bespoke mobile pyrolysis unit is set, transportation to the testing site will be undertaken followed by testing of the unit including emissions testing.

May 2024

Composting: NSW DPIRD commenced cattle carcass composting trials in November 2023 and pig carcass trials in February 2024.

Highlights:

- The cattle carcass composting trial is comparing current industry practice with 3 alternative composting treatments. All treatments have achieved the temperatures required for pasteurisation.
- A novel aspect of the trial includes the insertion of temperature loggers within the carcasses to determine if these areas of the composting process are reaching appropriate temperatures for the elimination of pathogens.
- The pig carcass composting trial compares standard whole carcass composting with the grinding of partially composted carcasses before completing the composting process. Initial microbiological testing showed that the material complies with the microbiological thresholds outlined in Australian Standard AS4454 (2012).

Pyrolysis: A handover date for the bespoke mobile pyrolysis unit has been set for the first half of 2024. Afterwards, transportation to the testing site will be undertaken followed by testing of the unit including emissions testing.

February 2024

Pyrolysis: Construction of the bespoke mobile pyrolysis unit is ongoing. Planning for the testing phase of the pyrolysis unit is underway including suitable site location and logistics. Testing protocol and positioning of the emissions testing ports within the stack of the unit have been completed with an emissions consultant.

Compositing: A third treatment commenced in January 2024 using double carcass layer per windrow as opposed to single layers for the other treatments, and the control.

Samples for microbial analysis were collected from the control and second treatment rounds. Temperature probes were inserted inside and next to the carcasses to compare internal carcass with compost pile temperatures. All the treatments, except for the control, reached temperatures required for pasteurisation (>55oC) within 24 hours of setup.

November 2023

Composting: A field trial was successfully completed at the Rivalea piggery in Corowa NSW. It provided important preliminary data on process performance and insight into how grinding and composting could be implemented effectively in an outbreak.

Two whole carcass composting treatments - the first using recommended carbon to nitrogen ratios, and the second using five turns to aerate the material - as well as a control treatment commenced in November 2023.

Source: https://www.agriculture.gov.au/agriculture-land/animal/health/animal-plan/project-32